

ELECTRIC FOUR-WHEELER DESIGN CHALLENGE

Project Report

SAE India Southern Section — EFWDC 2025
SASTRA University, Thanjavur — National Level

Team Name	Denji Squad
College	Knowledge Institute of Technology (KIOT)
Team ID	EFWDC2025001
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Achievement	Overall 3rd Place — National Level
Prepared by (Role)	Testing & Validation Engineer / Co-Driver

1. Abstract

This report presents the design and development of an electric four-wheeler built by Team Denji Squad (Knowledge Institute of Technology) for the SAE India Electric Four-Wheeler Design Challenge (EFWDC) 2025. The vehicle is a two-seater platform driven by dual BLDC hub motors and powered by a 48V lithium-based battery pack. The design process spanned hand-sketches and CAD layout modeling, Ackermann steering geometry, braking and suspension calculations, CAE structural analysis of the chassis, and a fully costed Engineering Bill of Materials. The team's prototype secured the **Overall 3rd Place** position at the National Level of the competition, held at SASTRA University, Thanjavur. This report also documents the author's contribution as Testing & Validation Engineer / Co-Driver.

2. Introduction

Electric mobility is an increasingly important area of automotive engineering, driven by the need for sustainable, low-emission transport. The EFWDC challenges student teams to design, simulate, and (in later stages) fabricate a working electric four-wheeler, applying core automotive engineering principles across chassis design, powertrain integration, suspension and steering geometry, braking systems, and cost engineering. This report documents Team Denji Squad's design approach as captured in their Design Report I submission.

3. Problem Statement

To design a lightweight, energy-efficient, and structurally sound electric four-wheeler that balances performance, safety, and cost, using validated engineering calculations and simulations rather than assumptions — while meeting SAE India's Design Report formatting and technical requirements.

4. Objectives

- Develop a two-seater electric four-wheeler layout via hand-sketch and CAD modeling.
- Calculate and validate critical dimensions: chassis frame sizing, Ackermann steering geometry, and suspension parameters.
- Simulate powertrain elements including BLDC motor drive and battery charging behavior.
- Perform CAE-based structural analysis (stress and deflection) on the chassis.
- Build a structured, traceable Engineering Bill of Materials (E-BOM) with cost estimation.
- Test and validate the vehicle's systems ahead of competition.

5. Project Overview

The vehicle was designed as a 2-seater electric four-wheeler with an overall length of 2500 mm, front width of 1300 mm, height of 1000 mm, and a wheelbase of 2000 mm. Design work progressed from hand-drawn graph-paper sketches (scale 1:5) through full-scale CAD modeling (scale 1:1), covering front, top, and side views. Key subsystems — battery pack, motor, steering column, and seating — were laid out and positioned within this envelope before detailed calculations were performed.

6. Electric Vehicle Architecture

The vehicle uses a dual BLDC hub-motor drive architecture (motors mounted at the rear wheels), powered by a 48V lithium-based battery pack through a motor controller. A three-phase inverter with Hall-effect sensor feedback and PID-based closed-loop control governs motor speed and torque delivery. The chassis is a mild-steel (MS) rectangular tube frame with a bamboo-frame element (as specified in the Bill of Materials), engineered to house the seating, steering column, and battery pack under the seating area.

7. Electrical System

Parameter	Value
System voltage	48 V
Motor current	45 A
Motor power	2 kW per motor × 2 = 4 kW total
BLDC motor efficiency	85%
Motor RPM (as stated in source doc)	64,000 RPM — flagged as likely erroneous / unverified
Torque (max / min, per source)	60 / 30 (units as listed in source; per-motor basis)
Motor controller	Steel-housed unit (BOM-confirmed)

Note on data conflicts: The source Design Report contains internally inconsistent figures for total motor power — stated variously as ~4 kW, 4.1 kW (arithmetic result of 2×2kW), and elsewhere as 4.8 kW (2×2.4kW). The 2 kW-per-motor figure (giving 4 kW total) is the most consistently repeated value across the document and is used as the primary reference here. Motor RPM is listed as 64,000 RPM in the source, which is inconsistent with typical hub-motor performance for this power class — this figure could not be independently verified and is flagged rather than corrected.

8. BLDC Motor Drive

The motor drive circuit is a three-phase inverter built from six switching devices (MOSFETs/IGBTs), powering the BLDC motor's three-phase terminals. Rotor position is sensed via Hall-effect sensors, which feed a commutation-logic block to determine correct switch states, translated into gate pulses by a dedicated gate-pulse block. Closed-loop speed control is implemented using a PID controller, comparing measured rotor speed against a reference value (1000 RPM used in the Simulink model) and adjusting motor torque accordingly. This subsystem was designed and validated in MATLAB/Simulink.

9. Battery System

Parameter	Value
Voltage	48 V
Capacity	45 Ah
Energy capacity	2.16 kWh
Discharge rate	45 Ah / 1h

Chemistry (per BOM)	Lithium-based (listed as "Lithium-Phosphorus" in source BOM)
BMS	Present
DC-DC Converter	Present
Charging Port	Present

Charging Circuit: The battery is charged using a Constant-Current Constant-Voltage (CCCV) protocol, designed and simulated in MATLAB/Simulink. Below 40% state of charge (SOC), the circuit operates in Constant Current mode with a PI controller regulating current via switching gate pulses. Above 40% SOC, the circuit switches to Constant Voltage mode until the battery is fully charged. This staged approach allows fast initial charging while protecting the battery from overcharge.

10. Vehicle Control System

Motor speed and torque are governed by a PID control loop referencing Hall-sensor rotor position feedback, as described in Section 8. This forms the core closed-loop control system for vehicle propulsion in the simulated model.

11. Testing and Validation

The team's documented work plan (Task 7: Testing) identifies two planned validation streams:

Performance Testing (acceleration, top speed, and range evaluation) and **Safety Testing** (crash tests and passenger protection evaluation). The Design Report I document primarily covers design-stage calculations and simulations; detailed test logs and measured results were not included in the reviewed document.

12. My Role and Responsibilities

Role: Testing & Validation Engineer / Co-Driver

As Testing & Validation Engineer, my responsibilities focused on supporting the verification of the vehicle's electrical and mechanical systems ahead of competition, and contributing as co-driver during vehicle trial runs. This role sits within the broader test-and-validation phase of the project's engineering workflow (see Section 11).

13. Safety Considerations

- CCCV charging protocol to prevent battery overcharge.
- Disc brake system with dedicated braking-force, stopping-distance, and torque calculations performed at the design stage.
- CAE-based structural validation of the chassis to assess deformation under load before fabrication.
- Planned crash/safety testing as part of the project's test phase (see Section 11).

14. Results / Achievement

Team Denji Squad's Electric Four-Wheeler Design Challenge entry achieved the **Overall 3rd Place** position at the National Level competition held at SASTRA University, Thanjavur. This reflects the

team's combined design, simulation, and engineering-documentation effort, of which testing and validation formed one contributing part.

15. Challenges

- Reconciling design assumptions across iterative calculation sections (e.g., differing mass and speed values used in different subsystem calculations).
- Achieving structural strength targets within chassis weight constraints.
- Managing battery energy budget (2.16 kWh) against total system power draw.
- Coordinating design and validation work across a large, multi-subsystem team.

16. Applications (Potential)

The following are potential applications of this electric four-wheeler concept — not implemented deployments:

- Last-mile personal or campus electric mobility.
- Low-speed urban commuting prototype platform.
- Educational platform for EV powertrain, controls, and vehicle dynamics learning.
- Base platform for further student EV research and development.

17. Future Scope

Proposed future improvements (not yet implemented):

- Improved battery management and thermal monitoring.
- Higher energy-efficiency drivetrain components.
- Advanced motor control tuning (e.g., field-oriented control).
- Vehicle telemetry and data-logging systems.
- Regenerative braking integration.
- Advanced safety monitoring systems.
- Refined suspension and chassis integration.

18. Conclusion

Team Denji Squad's Electric Four-Wheeler Design Challenge project demonstrated a structured, calculation-driven approach to EV design — spanning CAD layout, steering and suspension geometry, braking analysis, CAE structural validation, motor and charging-circuit simulation, and cost-engineered material selection. The project's Overall 3rd Place finish at the National Level reflects this rigor. As Testing & Validation Engineer / Co-Driver, my contribution supported the verification stage of this broader engineering effort. The foundation established here provides a strong basis for future refinement of the vehicle's electrical and mechanical systems.

19. References

- Team Denji Squad, "Electric Four-Wheeler Design Challenge — 2025, Design Report I," SAE India Southern Section, Knowledge Institute of Technology, Team ID EFWDC2025001.

- SAE India Southern Section — Electric Four-Wheeler Design Challenge (EFWDC) 2025 competition guidelines.

This report was compiled from the team's Design Report I document and information provided by the author. Where source data contained internal inconsistencies, this has been noted rather than resolved by assumption.